Tutorial - 12

Bootrap → Sampling with Replacement

sample:	Feature 1 (Age)	feature 2 (income)	feature 3 (credit score)	Target (Loan Approved)
S_1	25	40	650	Yes
S_2	30	60	700	Yes
S_3	45	33	600	No
S_4	50	30	580	No
S_5	28	55	720	Yes

Tasks:

1. Create 3 boostrap samples (size = 5) from the dataset.
2. Train 3 decision tree models using each bootrap sample
3. for a new sample (Age = 32, Income = 50, credit score = 690) determine predictions from all trees.
4. Compute the final prediction using majority voting.

1) Boostrap 1 (B_1) S_1, S_2, S_2, S_4, S_5

Boostrap 2 (B_2) S_2, S_3, S_4, S_4, S_5

Bootsrap 3 (B_3) S_1, S_2, S_3, S_3, S_4

Decision Tree (B_1) Bootstrap 2 (B_2) Bootstrap 3 (B_3)

B_1	B_2	B_3
S_1, S_2, S_2, S_4, S_5	S_2, S_3, S_4, S_4, S_5	S_1, S_2, S_3, S_3, S_4
Y Y Y N Y	Y N N N Y	Y Y N N N

Age
≤ 30 / \ > 30
Yes No

credit score
/ ≥ 650 \ < 650
Yes NO

Income
≥ 40 / \ < 40
Yes NO

=> single split DT

3&4 prediction & voting

Age = 32
↓
loan Approval : N

Income = 50
↓
loan Approval : Y

Credit score = 690
↓
loan Approval : Y

majority voting

N Y Y (Y)

Age = 35
↓
loan approval : N

Income = 45
↓
Y

Credit score = 750
↓
Y

majority voting

N Y Y (Y)